

Academic Year
2022-23
V SEM CSE
Handbook
(V20 Regulation)

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INSTITUTE

A diagram consisting of a rectangular box with a slanted top-left corner. Inside the box is the word 'INSTITUTE' in a purple, serif, all-caps font. A vertical line extends from the bottom center of the box, ending in a downward-pointing arrowhead. This arrowhead points to a horizontal banner with wavy, ribbon-like edges. Inside the banner are the words 'VISION MISSION' in the same purple, serif, all-caps font.

VISION MISSION

INSTITUTE VISION AND MISSION

VISION

To be a premier technological institute striving for excellence with global perspective and commitment to the nation.

MISSION

- To produce engineering graduates of professional quality and global perspective through Learner Centric Education.
- To establish linkages with government, industry and research laboratories to promote R&D activities and to disseminate innovations.
- To create an eco-system in the institute that leads to holistic development and ability for life-long learning.

DEPARTMENT

VISION

MISSION

DEPARTMENT

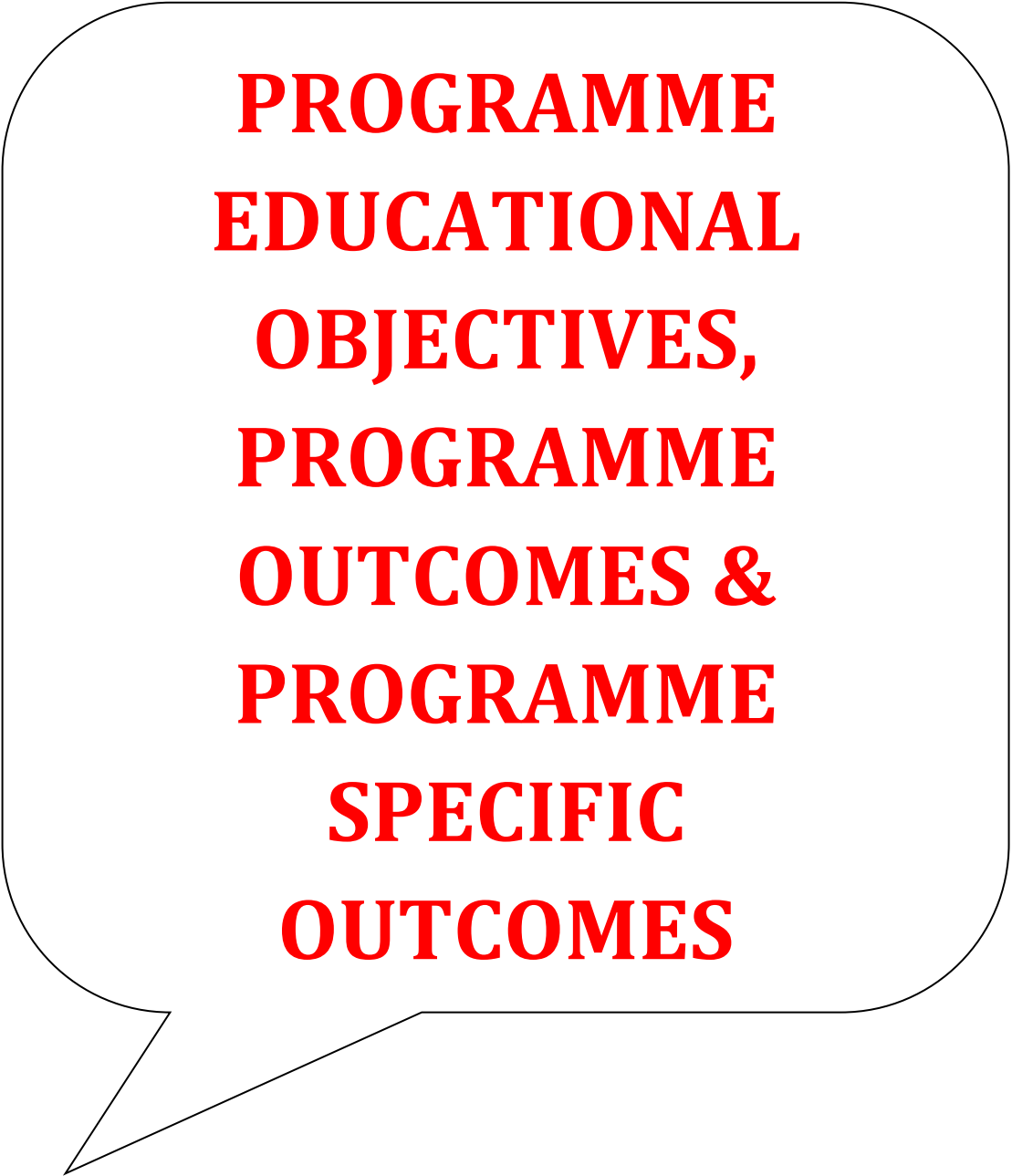
VISION AND MISSION

Vision:

- To evolve as a centre of academic and research excellence in the area of Computer Science and Engineering.

Mission :

- To utilize innovative learning methods for academic improvement.
- To encourage higher studies and research to meet the futuristic requirements of Computer Science and Engineering.
- To inculcate Ethics and Human values for developing students with good character



**PROGRAMME
EDUCATIONAL
OBJECTIVES,
PROGRAMME
OUTCOMES &
PROGRAMME
SPECIFIC
OUTCOMES**

Programme Educational Objectives (PEOs)

Graduates of this programme will :

PEO 1: Adapt to evolving technology.

PEO 2: Provide optimal solutions to real time problems.

PEO 3: Demonstrate his/her abilities to support service activities with due consideration for Professional and Ethical Values.

Programme Specific Outcomes (PSO s):

A graduate of the Computer Science and Engineering Program will be able to:

PSO 1: Use Mathematical Abstractions and Algorithmic Design along with Open Source Programming tools to solve complexities involved in Programming. [K3]

PSO 2: Use Professional engineering practices and strategies for development and maintenance of software. [K3]

Programme Outcomes (POs):

Computer Science Engineering Graduates will be able to:

1. **Engineering knowledge:** Apply the knowledge of Mathematics, Science, Engineering Fundamentals and Concepts of Computer Science Engineering to the solution of complex Engineering problems. [K3]
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of Mathematics, Natural Sciences and Computer Science. [K4]
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specific needs with appropriate consideration for the public health and safety, and the cultural, societal and environmental considerations. [K5]
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions. [K5]
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex Engineering activities with an understanding of the limitations. [K3]
6. **The Engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional Engineering practice. [K3]
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development. [K3]
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the Engineering practice. [K3]
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings. [K6]
10. **Communication:** Communicate effectively on complex Engineering activities with the Engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions. [K2]
11. **Project management and finance:** Demonstrate knowledge and understanding of the Engineering and Management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments. [K6]
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change. [K1]

ACADEMIC CALENDAR

✉ : principal@srivasaviengg.ac.in
svec.a8@gmail.com



☎ : 08818- 284344, 355

SRI VASAVI ENGINEERING COLLEGE (AUTONOMOUS)

(Sponsored by Sri Vasavi Educational Society)

(Approved by AICTE, New Delhi & Permanently affiliated to JNTUK, Kakinada)

(Accredited by NAAC with 'A' Grade, Recognized by UGC under section 2(f) & 12(B))

(NBA Accreditation to B.Tech., EEE, CSE, ME and ECE Branches for 3 Years)

Pedatadepalli, **TADEPALLIGUDEM – 534 101**, W.G.Dist. (A.P)

Principal's Office

Date: 02-08-2022

Academic Calendar

For B.Tech V and VI Semesters , Academic Year 2022-23

V Semester			
Description	From	To	Weeks
Commencement of Class Work	08.08.2022		
I Unit of Instructions	08.08.2022	24.09.2022	7 W
I Mid Examinations	26.09.2022	01.10.2022	1 W
II Unit of Instructions	03.10.2022	19.11.2022	7 W
II Mid Examinations	21.11.2022	26.11.2022	1 W
Preparation and Practicals	28.11.2022	03.12.2022	1 W
End Examinations	05.12.2022	16.12.2022	2 W
Commencement of VI Semester Class Work	19.12.2022		
VI Semester			
I Unit of Instructions	19.12.2022	04.02.2023	7 W
I Mid Examinations	06.02.2023	11.02.2023	1 W
II Unit of Instructions	13.02.2023	01.04.2023	7 W
II Mid Examinations	03.04.2023	08.04.2023	1 W
Preparation and Practicals	10.04.2023	15.04.2023	1 W
End Examinations	17.04.2023	28.04.2023	1 W
Next Semester Class Work	12.06.2023		

PRINCIPAL

Copy to: ALL

Vision

To be a premier technological institute striving for excellence with global perspective and commitment to the nation.

Mission

- To produce Engineering graduates of professional quality and global perspective through learner-centric education.
- To establish linkages with government, industry and Research laboratories to promote R&D activities and to disseminate innovations.
- To create an eco-system in the institute that leads to holistic development and ability for life-long learning.



SRI VASAVI ENGINEERING COLLEGE (Autonomous)

Pedatadepalli, TADEPALLIGUDEM-534 101, W.G. Dist.

Department Of Computer Science & Engineering (Accredited by NBA)



CLASS CONSOLIDATED TIME TABLE

Class: V Semester

w.e.f. 10.08.2022

Section: A

Class Coordinator: Mrs.G.NagaVallika

Room No : PG-201

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time Day	(09.30 AM- 10.30 AM)	(10.30 AM- 11.20 AM)	(11.20 AM- 12.10 PM)	(12.10 PM- 01.00 PM)		(02.00 PM- 02.50 PM)	(02.50 PM- 03.40 PM)	(03.40 PM- 04.30 PM)
Mon	PCS-III(V)		DM	OS	Lunch Break	PCS-III(A)		WT
Tue	DM	OS	AI	WT		WTLAB		
Wed	OS	MCCP-I (JOE)				WT	AI	DM
Thu	WT	AI	OS	DM		MCCP-I (JOE)		
Fri	AI	DM LAB				DM	DS	LIBRARY
Sat	WT	AI	WT	AI		DS	DM	SPORTS

Section: B

Class Coordinator: Mrs.R.Padmaja

Room No : PG-202

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time Day	(09.30 AM- 10.30 AM)	(10.30 AM- 11.20 AM)	(11.20 AM- 12.10 PM)	(12.10 PM- 01.00 PM)		(02.00 PM- 02.50 PM)	(02.50 PM- 03.40 PM)	(03.40 PM- 04.30 PM)
Mon	WT	OS	PCS-III(A)		Lunch Break	AI	DM	LIBRARY
Tue	AI	WT	DM	OS		MCCP-I (JOE)		
Wed	PCS-III(V)		WT	AI		DM	OS	SPORTS
Thu	OS	DM	WT	AI		WT LAB		
Fri	WT	MCCP-I (JOE)				OS	AI	DM
Sat	DM	DM LAB				AI	WT	OS

Section: C

Class Coordinator: Ms. D.SasiRekha

Room No : PG-203

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time Day	(09.30 AM- 10.30 AM)	(10.30 AM- 11.20 AM)	(11.20 AM- 12.10 PM)	(12.10 PM- 01.00 PM)		(02.00 PM- 02.50 PM)	(02.50 PM- 03.40 PM)	(03.40 PM- 04.30 PM)
Mon	DM	WT LAB			Lunch Break	OS	WT	OS
Tue	PCS-III(A)		PCS-III(V)			DM	WT	AI
Wed	OS	AI	DM	AI		DM LAB		
Thu	WT	MCCP-I (JOE)				AI	DM	OS
Fri	AI	AI	WT	LIBRARY		OS	WT	SPORTS
Sat	WT	OS	DM	DM		MCCP-I (JOE)		

Section: D

Class Coordinator: Mr. K.Praveen kumar

Room No : PG-204

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time Day	(09.30 AM- 10.30 AM)	(10.30 AM- 11.20 AM)	(11.20 AM- 12.10 PM)	(12.10 PM- 01.00 PM)		(02.00 PM- 02.50 PM)	(02.50 PM- 03.40 PM)	(03.40 PM- 04.30 PM)
Mon	OS	MCCP-I (JOE)			Lunch Break	PCS-III(V)		AI
Tue	WT	OS	PCS-III(A)			AI	DM	WT
Wed	DM	WT LAB				OS	WT	SPORTS
Thu	AI	DM	WT	AI		OS	WT	LIBRARY
Fri	WT	DM	DM	OS		DM LAB		
Sat	AI	MCCP-I (JOE)				AI	DM	OS

Staff Details:

S. No.	Course Code	Course Name	Section			
			A	B	C	D
1.	V20CST10	Operating Systems	Mrs.Y.Divya Vani	Mrs.Y.Divya Vani	Ms.D.Sasi Rekha	Ms.D.Sasi Rekha
2.	V20CST11	Data Mining	Mrs.N.Hiranmayee	Mrs.N.Hiranmayee	Dr.K.SrinivasaRao	Dr.K.SrinivasaRao
3.	V20CST12	Web Technologies	Mrs.R.Padmaja	Mrs.R.Padmaja	Mr.K.Praveen Kumar	Mr.K.Praveen Kumar
4.	V20CSTJE01	Job Oriented Elective: Master Coding in Competitive Programming	Dr. V.Venkateswara Rao/ Ms.S.S.L.Surekha	Dr. V.Venkateswara Rao/ s.S.S.L.Surekha	Dr. V.Venkateswara Rao/ Ms.S.S.L.Surekha	Dr. V.Venkateswara Rao/ Ms.S.S.L.Surekha
5.	V20CSTPE03	Elective-I : Artificial Intelligence	Ms.G.Nagavallika	Ms.G.Nagavallika	Dr.G.Loshma	Dr.G.Loshma
6.	V20CSL09	Data Mining Lab	Mrs.N.Hiranmayee Mrs.Y.Divya Vani	Mrs.N.Hiranmayee Mrs.Y.Divya Vani	Dr.K.SrinivasaRao Dr.G.Loshma	Dr.K.SrinivasaRao/ Dr.G.Loshma
7.	V20CSL10	Web Technologies Lab	Ms.R.Padmaja Ms.D.SasiRekha	Ms.R.Padmaja Ms.G.Nagavallika	Mr.K.Praveen Kumar/ Ms.G.Nagavallika	Mr.K.Praveen Kumar/ Mr.M.Y.Sekharam
8.	V20ENT03	Professional Communication Skills-III	Mrs.U.Aparanji/ Mr.J.N.V.Somayajulu	Dr.T.Sujani/ Mr.P.Someswara Rao	Mr.K.V.RamaRao/ Mr.J.N.V.Somaya julu	Mrs.K.Radha Madhavi/ Mr.P.Someswara Rao

Lab Venues:

S.No.	Name of the Lab Course	Lab Venue
1	Data Mining Lab	LINUS TORVALDS LAB
2	Web Technologies Lab	
3	Job Oriented Elective: Master Coding in Competitive Programming	YELLOW LAB



Head of the Department

COURSE STRUCTURE

V Semester

S.No.	VII – Semester						
	Course Code		Course	L	T	P	C
1	V20CST10	PCC	Operating Systems	3	0	0	3
2	V20CST11	PCC	Data Mining	3	0	0	3
3	V20CST12	PCC	Web Technologies	3	0	0	3
4	V20CSTJE01	OEC	Job Oriented Elective-I	3	0	0	3
		JOE	Mastering Coding and Competitive Programming - Part-1	0	0	6	3
5	Professional Elective-I						
	V20CSTPE01	PEC	1. Software Testing Methodologies	3	0	0	3
	V20CSTPE02		2. Principles of Programming Languages				
	V20CSTPE03		3. Artificial Intelligence				
	V20CSTPE04		4. Computer Graphics				
6	V20CSL09	PCC	Data Mining Lab	0	0	3	1.5
7	V20CSL10	PCC	Web Technologies Lab	0	0	3	1.5
8	V20SOC03	SO/SS	Skill Oriented Course-III (Soft Skills)	1	0	2	2
9	V20CSP01	Internship	Mini Project / Internship	0	0	3	1.5
10	V20ENT04	MNC	Professional Communication Skills -III	2	0	0	0
Total				16	0	11	21.5



LESSON PLANS

Operating Systems

Academic Year: 2022-23

Year/ Semester: V

Name of the Course: Operating Systems

Programme: B.Tech

Section: A,B,C& D

Course Code:V20CST10

LESSON PLAN

Course Outcomes (Along with Knowledge Level):

After completion of this course, Student will be able to:

S.No.	CO No.	Course Outcome	BTL
1.	CO1	Describe Operating System Services and System Calls	K2
2.	CO2	Illustrate Process Management Concepts and CPU Scheduling Algorithms	K3
3.	CO3	Demonstrate Process Synchronization primitives and Process Deadlocks	K3
4.	CO4	Illustrate Memory Management Techniques and Page Replacement Algorithms	K3
5.	CO5	Describe File System Concepts and Mass Storage Structures	K2

Text Books:

1. Operating System Concepts, AbrahamSilberschatz, ,Peter Baer Galvin,Greg Gagne, 9th Edition, John Wiley and Sons Inc., 2012

Reference Books:

1. Operating Systems – Internals and Design Principles, William Stallings, 7th Edition, Prentice Hall, 2012
2. Modern Operating Systems, Andrew S. Tanenbaum, Third Edition, Addison Wesley,2007

Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5
Targeted Proficiency Level		65	60	60	60	65
Targeted level of Attainment	Level 3	65	60	60	60	65
	Level 2	55	50	50	50	55
	Level 1	45	40	40	40	45

Lecture Plan:

SNo	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	C01	Dissemination of Department Vision, Mission, PEOs, POs, PSOs	-	-	-	
		Introduction of OS	K2	1	Lecture with Discussion	BB/ICT
		Operating-System Structure	K2	1	Lecture with Discussion	BB/ICT
		Operating-System Services	K2	1	Lecture with Discussion	BB/ICT
		User and Operating-System Interface	K2	1	Lecture with Discussion	BB/ICT
		System Calls	K2	1	Lecture with Discussion	BB/ICT
		Types of System Calls	K2	1	Lecture with Discussion and in class assignment	BB/ICT
Total				6		
2	C02	Process Concept and Process Scheduling	K3	2	Lecture with Discussion	BB/ICT
		Operations On Processes	K3	1	Lecture with Discussion	BB/ICT
		Inter Process Communication	K3	1	Lecture with Discussion	BB/ICT
		Threads overview	K3	1	Lecture with Discussion	BB/ICT
		Multithreading Models	K3	1	Lecture with Discussion	BB/ICT
		CPU Scheduling Basic Concepts and CPU Scheduling Criteria	K3	1	Lecture with Discussion	BB/ICT
		CPU Scheduling Algorithms	K3	3	Lecture with Discussion and in class Assignment	BB/ICT
Total				10		

3	C03	Critical Section Problem	K3	1	Lecture with Discussion	BB/ICT
		Peterson's Solution	K3	1	Lecture with Discussion	BB/ICT
		Synchronization Hardware	K3	1	Lecture with Discussion	BB/ICT
		Mutex Locks	K3	1	Lecture with Discussion	BB/ICT
		Semaphores	K3	1	Lecture with Discussion and in class Assignment	BB/ICT
		Classic Problems of Synchronization	K3	2	Lecture with Discussion and in class Assignment	BB/ICT
		Monitors	K3	1	Lecture with Discussion	BB/ICT
		System Model and DeadLock Characterization	K3	1	Lecture with Discussion	BB/ICT
		Methods for Handling Deadlocks	K3	1	Lecture with Discussion	BB/ICT
		Deadlock Prevention	K3	1	Lecture with Discussion	BB/ICT
		Deadlock Avoidance	K3	1	Lecture with Discussion and in class Assignment	BB/ICT
		Deadlock Detection	K3	1	Lecture with Discussion	BB/ICT
		Recovery from Deadlock	K3	1	Lecture with Discussion	BB/ICT
		Total		14		
4	C04	Swapping and Contiguous Memory Allocation	K3	1	Lecture with Discussion	BB/ICT
		Segmentation	K3	1	Lecture with Discussion	BB/ICT

		Paging	K3	2	Lecture with Discussion and in class Assignment	BB/ICT
		Structure of the Page Table	K3	1	Lecture with Discussion	BB/ICT
		Demand Paging	K3	1	Lecture with Discussion	BB/ICT
		Page Replacement Algorithms	K3	2	Lecture with Discussion and in class Assignment	BB/ICT
		Allocation of Frames	K3	1	Lecture with Discussion	BB/ICT
		Thrashing	K3	1	Lecture with Discussion	BB/ICT
		Total			10	
5	C05	Overview of Mass-Storage Structure	K2	1	Lecture with Discussion	BB/ICT
		Disk Scheduling	K2	2	Lecture with Discussion and in class assignment	BB/ICT
		File Concept, Access Methods	K2	1	Lecture with Discussion	BB/ICT
		Directory and Disk Structure	K2	1	Lecture with Discussion	BB/ICT
		File-System Mounting	K2	1	Lecture with Discussion	BB/ICT
		File Allocation Methods	K2	1	Lecture with Discussion	BB/ICT
Total			7			

Total No. of Classes: 47

*BB-Black Board

*ICT- Information & Communication Technology

Data Mining

Academic Year: 2022-23

Year/ Semester: V

Name of the Course: Data Mining

Programme: B.Tech

Section: A,B,C& D

Course Code:V20CST11

LESSON PLAN

Course Outcomes (Along with Knowledge Level):

After completion of this course, Student will be able to:

S. No.	CO No.	Course Outcome	BTL
1.	CO1	Explain the concept of Data Mining and its functionalities.	K2
2.	CO2	Discuss various Data Preprocessing Techniques.	K2
3.	CO3	Demonstrate Association Analysis Techniques.	K3
4.	CO4	Illustrate various Classification Techniques.	K3
5.	CO5	Use different Clustering techniques to cluster data.	K3

Text Books:

1. Data Mining Concepts and Techniques, Jiawei Han, Micheline Kamber, Jian Pei, 3rd Edition, Morgan Kaufmann Publishers

Reference Books:

1. Introduction to Data Mining, Pang-Ning Tan, Michael Steinbach, Vipin Kumar, 1st Edition, Pearson Education Inc.

2. Data Mining and Analysis, Mohammed J Zaki, Wagner Meira JR, 1st Edition, Cambridge University Press

Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5
Targeted Proficiency Level		60	60	60	60	60
Targeted level of Attainment	Level 3	65	65	60	60	60
	Level 2	55	55	50	50	50
	Level 1	45	45	40	40	40

Lecture Plan: UNIT-1

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1	CO1	Dissemination of Department Vision, Mission, PEOs, POs, PSOs, COs, Introduction: Identify what motivated Data Mining.	K1	1	Lecture	BB/ICT
2		State the importance of Data Mining. Describe kinds of Data on which Data Mining can be done.	K1	2	Lecture	BB/ICT
3		Illustrate Data Mining Functionalities.	K2	2	Lecture+ discussion	BB/ICT
4		Illustrate Major Issues in Data Mining	K1	1	Lecture	BB/ICT
5		Explain Attribute Types and Basic Statistical Descriptions of Data	K2	2	Lecture + discussion	BB/ICT
6		Illustrate Data Visualization techniques.	K2	1	Lecture + discussion	BB/ICT
7		Describe Data Similarity and Dissimilarity measures	K2	2	Lecture + discussion	BB/ICT
Total				11		

UNIT-2

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1	CO2	Data Pre-processing: Identify reasons for pre-processing the data.	K1	1	Lecture	BB/ICT
2		Describe Descriptive Data Summarization.	K2	1	Lecture + discussion	BB/ICT
3		Explain Data Cleaning techniques.	K2	2	Lecture + discussion	BB/ICT
4		Discuss Data Integration and Data Transformation techniques	K2	2	Lecture + discussion	BB/ICT
5		Explain Data Reduction techniques.	K2	2	Lecture + discussion	BB/ICT
6		Discuss Data Discretization	K2	2	Lecture +	BB/ICT

		and Concept Hierarchy Generation.			discussion	
Total				10		

UNIT-3

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1	CO3	Illustrate the concept of Association Analysis	K2	2	Lecture + discussion	BB/ICT
2		Explain Frequent Item set generation process.	K2	2	Lecture + discussion	BB/ICT
3		Discover Association Rules from the given data.	K3	2	Lecture + discussion and In-class Assignment	BB/ICT
4		Present a Compact representation of frequent item sets.	K1	1	Lecture	BB/ICT
5		Use FP-Growth Algorithm for Association Analysis.	K3	2	Lecture + discussion and In-class Assignment	BB/ICT
Total				09		

UNIT-4

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1	CO 4	Classification : Describe the basic concepts of Classification	K1	1	Lecture	BB/ICT
2		Illustrate the approach for solving a classification problem.	K2	1	Lecture + discussion	BB/ICT
3		Explain the concept of a Decision Tree Induction	K2	2	Lecture + discussion	BB/ICT
4		Construct a decision tree.	K3	1	Lecture + discussion and In-class Assignment	BB/ICT

5		Describe the methods for expressing an attribute test conditions.	K2	1	Lecture + discussion	BB/ICT
6		Identify the measures for selecting the best split.	K2	1	Lecture + discussion	BB/ICT
7		Illustrate Tree Pruning	K3	2	Lecture + discussion and In-class Assignment	BB/ICT
Total				9		

UNIT-5

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1	CO5	Explain the Concept of Cluster Analysis.	K2	1	Lecture + discussion	BB/ICT
2		Describe different types of Clustering and Different Types of Clusters.	K2	2	Lecture + discussion	BB/ICT
3		Apply k-means and its variants for clustering.	K3	3	Lecture + discussion and In-class Assignment	BB/ICT
4		Describe k-means additional issues.	K1	1	Lecture	BB/ICT
5		Describe k-means and different types of clusters.	K2	1	Lecture + discussion	BB/ICT
6		Identify k-means strengths and weaknesses	K1	1	Lecture	BB/ICT
7		Express k-means as an Optimization Problem.	K2	1	Lecture + discussion	BB/ICT
8		Apply Basic Agglomerative Hierarchical Clustering.	K3	1	Lecture + discussion and In-class Assignment	BB/ICT
9		Discuss other Specific Techniques for Agglomerative Hierarchical Clustering	K2	1	Lecture + discussion	BB/ICT

10		Explain Traditional Density: Center-Based Approach.	K2	1	Lecture + discussion	BB/ICT
11		Apply the DBSCAN Algorithm.	K3	1	Lecture + discussion and In-class Assignment	BB/ICT
12		List strengths and Weaknesses of DBSCAN Algorithm	K1	1	Lecture	BB/ICT
Total				15		

Total No. of Classes: 54

*BB-Black Board

*ICT- Information & Communication Technology

Web Technologies

Academic Year: 2022-23

Year/ Semester: V

Name of the Course: Web Technologies

Programme: B.Tech

Section: A,B,C& D

Course Code: V20CST12

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level):

After completion of this course, the students will be able to:

S.No.	CO No.	Course Outcome	BTL
1.	CO 1	Illustrate the basic concepts of HTML and CSS.	K2
2.	CO 2	Illustrate Extensible markup language and XML parsers	K3
3.	CO 3	Develop web applications using JDBC.	K3
4.	CO 4	Build database driven web applications using JSP	K3
5.	CO 5	Illustrate the basic concepts of Angular and NODE JS.	K2

Text Books:

1. Programming the World Wide Web, 7th Edition, Robert W Sebesta, Pearson, 2013.
2. Node.js, MongoDB and Angular Web Development, 2nd Edition, Brad Dayley, Brendan Dayley, Caleb Dayley, Pearson Education, 2018
3. JSP: The Complete reference, Phil Hanna, The McGraw-Hill Companies, 2001.
4. JDBC, Servlets, and JSP, New Edition, Santhosh Kumar K, Kogent Learning Solutions Inc, Dreamtech Press, 2018.

Reference Books:

1. Web Technologies: HTML, JavaScript, PHP, Java, JSP, XML and AJAX, Black book, Dream Tech.
2. An Introduction to Web Design, Programming, Paul S Wang, Sanda S Katila, Cengage Learning
3. Web Technologies, 1st Edition 7th impression, Uttam K Roy, Oxford, 2012.

Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5
Targeted Proficiency Level		60	60	60	60	60
Targeted level of Attainment	Level 3	60	60	60	60	60
	Level 2	55	55	55	55	55
	Level 1	50	50	50	50	50

Lecture Plan:

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1	C01	Describe Basic Syntax, Standard HTML Document Structure	K1	1	Lecture	BB
2		Describe Basic HTML Tags	K1	1	Lecture	BB+ICT
3		Explain Lists, Tables, Images	K2	2	Lecture	BB+ICT
4		Explain Forms, Frames	K2	2	Lecture	BB+ICT
5		Discuss Cascading style sheets, levels of style sheets and its formats	K2	2	Lecture	BB+ICT
6		List Properties of Cascading style sheets	K1	2	Lecture	BB+ICT
7		Explain Selector Forms	K2	2	Lecture	BB+ICT
8		Explain List Properties, Color and alignment of text	K2	3	Lecture	BB+ICT
Total				15		
1	C02	Explain of XML , Syntax, XML Document Structure	K2	3	Lecture	BB+ICT
2		Explain XML, Document type Definition, XML schemas, Namespaces	K2	3	Lecture	BB
3		Demonstrate Document object model, XSLT,	K3	3	Lecture with Discussion	BB+ICT
4		Illustrate DOM and SAX parsers	K3	3	Lecture with Discussion	BB+ICT
Total				12		
1	CO 3	Define about JDBC	K1	1	Lecture	BB+ICT
2		Describe about ODBC	K1	1	Lecture	BB+ICT
3		Explain about JDBC	K1	1	Lecture	BB+ICT
4		Explain the Components of JDBC	K1	2	Lecture	BB+ICT
5		Explain about JDBC Architecture	K1	1	Lecture	BB
6		Explain about Types of Drivers	K1	1	Lecture	BB
7		Demonstrate JDBC APIs	K3	3	Lecture	ICT
8		Define a Simple Application with JDBC	K1	2	Lecture	ICT
9		Explain about how to work prepared Statement	K2	2	Lecture	BB+ICT
Total				14		
1		Explain Servlets, Life cycle of		2	Lecture	BB+ICT

		Servlet,	K2			
2		List the Limitations of servlets, JSP Overview,	K1	2	Lecture	BB
3		Explain Components of a JSP Page: Directives, comments, Expressions, Scriptlets ,	K2	3	Lecture	BB
4	CO 4	Explain Components of a JSP Page: Declarations, implicit objects, Database Access, session tracking	K2	4	Lecture	BB+ICT
Total				11		
1		Demonstrate installing Node JS	K3	1	Lecture	BB+ICT
2		Explain Working with Node Packages.	K2	1	Lecture	BB
3		Demonstrate with Node JS Application	K2	2	Lecture	BB+ICT
4	C05	Understand of Angular, Modules, Directives,	K1	2	Lecture with Discussion	BB+ICT
5		Explain Data Binding, Dependency Injection, Services	K2	3	Lecture	BB+ICT
6		Explain about Separation of Responsibilities, Creating a Basic Angular Application.	K3	3	Lecture with Discussion	BB+ICT
Total				12		

Total No. of classes: 64

Mastering Coding and Competitive Programming

Academic Year: 2022-23

Programme: B.Tech

Year/ Semester: V

Section: A,B,C& D

Name of the Course: Mastering Coding and Competitive Programming

Course Code: V20CSTJE01

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level):

After completion of this course, the students will be able to:

S.No.	CO No.	Course Outcome	BTL
1.	CO1	Apply Mathematical reasoning and number theory to solve real world problems in linear time.	[K3]
2.	CO2	use of modular arithmetic, to solve complex problems in linear time , logarithmic.	[K3]
3.	CO3	use of Prime Factorization and complex solve problems.	[K3]
4.	CO4	Analyse different techniques including sieve to find prime numbers and evaluate efficiency of these methods.	[K4]
5.	CO5	Experiment with Hashing and searching techniques to solve problems on Arrays in Linear time.	[K3]

Text Books:

1. Java The Complete Reference – Eleventh Edition, Herbert Schildt, Oracle.
2. Guide to Competitive Programming by Antti Laaksonen
3. Programming Challenges by Steven's Skiena

Tools:

1. practice.geeksforgeeks.com
2. leetcode.com
3. codingninjas.com
4. hackerrank.com
5. interviewbit.com

Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5
Targeted Proficiency Level		80%	80%	80%	80%	80%
Targeted level of Attainment	Level 3	80%	80%	80%	80%	80%
	Level 2	70%	70%	70%	70%	70%
	Level 1	60%	60%	60%	60%	60%

Lecture Plan:

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO1	Develop Programs to solve problems based on Mathematical logic, Reasoning and number theory	K3	10	Lecture With Discussion	ICT
2	CO2	Develop programs using different techniques to find prime number	K3	10	Lecture With Discussion	ICT
3	CO3	Develop programs using Sieve method and optimize Complexity of finding prime number	K3	12	Lecture With Discussion	ICT
4	CO4	Develop programs based on series, patterns and	K4	16	Lecture With Discussion	ICT
		Develop programs on concept of Fibonacci series	K3			
		Develop programs on strings including palindrome and anagram concepts	K3			
5	CO5	Develop programs to search pattern in a string	K3	12	Lecture With Discussion	ICT
		Demonstrate DDL, DML commands and queries	K2			

Total No. of Classes: 60

Artificial Intelligence

Academic Year: 2022-23

Year/ Semester: V

Name of the Course: Artificial Intelligence

Programme: B.Tech

Section: A,B,C& D

Course Code: V20CSTPE03

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level):

After completion of this course, the students will be able to:

S. No.	CO No.	Course Outcome	BTL
1	CO1	Discuss the foundations of AI.	K2
2	CO2	Identify Search Strategies for Problem Solving.	K2
3	CO3	Illustrate Adversarial Search for Game Playing.	K2
4	CO4	Discuss Reasoning approaches.	K2
5	CO5	Illustrate Knowledge Representation approaches.	K2

Text Books:

1. Artificial Intelligence : A Modern Approach, Stuart J. Russell and Peter Norvig, 3rd Edition, PrenticeHall.

2. Artificial Intelligence, Elaine Rich, Kevin Knight, Shivashankar B Nair, 3rd Edition, Tata McGraw-Hill.

Reference Books:

1. Artificial Intelligence, George F Luger, Pearson Education Publications.

2. Artificial Intelligence, Saroj Kaushik, 1st Edition, Cengage Learning.

Proficiency and Attainment Levels for Course Outcomes in Percentages

Cos		CO1	CO2	CO3	CO4	CO5
Targeted Proficiency Level		60	60	60	60	60
Targeted level of Attainment	Level 3	60	60	60	60	60
	Level 2	50	50	50	50	50
	Level 1	40	40	40	40	40

Lecture Plan:

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO1	Dissemination of Department Vision, Mission, PEOs, POs, PSOs	-	-	-	
		Explain AI Introduction	K2	1	Lecture	ICT
2		Explain what is AI	K2	1	Lecture With Discussion	ICT
3		Discuss foundations of AI	K2	2	Lecture With Discussion	ICT
4		Discuss history of AI	K2	2	Lecture With Discussion	ICT
5		Discuss the state of art of applications	K2	1	Lecture With Discussion	ICT
6		Describe agents, environments	K2	1	Lecture With Discussion	ICT
7		Discuss the concept of rationality	K2	1	Lecture With Discussion	ICT
8		Discuss the nature of environments	K2	1	Lecture With Discussion	ICT
9		Discuss the Structure of agents	K2	2	Lecture With Discussion	ICT
Total				12		

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO2	Discuss Problem solving agents	K2	2	Lecture with Discussion and in class Assignment	ICT
2		Illustrate Example problems	K2	2	Lecture with Discussion and in class Assignment	ICT
3		Illustrate Searching for solutions	K2	2	Lecture with Discussion	ICT

					and in class Assignment	
4		Explain Uniformed search strategies	K2	2	Lecture with Discussion and in class Assignment	ICT
5		Explain Informed(Heuristic) search strategies	K2	2	Lecture with Discussion and in class Assignment	ICT
6		Explain Heuristic functions	K2	2	Lecture with Discussion and in class Assignment	ICT
7		Discuss Local search algorithms and optimization problems	K2	2	Lecture with Discussion and in class Assignment	ICT
Total				14		

S.N o	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO3	Explain Games	K2	1	Lecture with Discussion	BB/ICT
2		Discuss Optimal decisions in games	K2	1	Lecture with Discussion	BB/ICT
3		Discuss Alpha-Beta pruning	K2	1	Lecture with Discussion	BB/ICT
4		Describe Imperfect real time decisions	K2	1	Lecture with Discussion	BB/ICT
5		Describe Stochastic games	K2	1	Lecture with Discussion	BB/ICT
6		Describe Partially observable games	K2	2	Lecture with Discussion	BB/ICT
7		Explain State of art of game programs	K2	2	Lecture with Discussion	BB/ICT

8		Discuss Alternative approaches	K2	1	Lecture with Discussion	BB/ICT
Total				10		

S.N o	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO4	Explain Propositional Logic	K2	1	Lecture with Discussion	BB/ICT
2		Discuss Propositional Theorem proving	K2	2	Lecture with Discussion	BB/ICT
3		Discuss Syntax and Semantics of First order logic	K2	2	Lecture with Discussion	BB/ICT
4		Explain first order logic	K2	2	Lecture with Discussion	BB/ICT
5		Describe Forward chaining	K2	1	Lecture with Discussion	BB/ICT
6		Describe Backward chaining	K2	1	Lecture with Discussion	BB/ICT
7		Discuss Resolution	K2	2	Lecture with Discussion	BB/ICT
Total				11		

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO5	Explain Knowledge representation and mapping	K2	2	Lecture with Discussion	BB/ICT
2		Discuss the Approaches to knowledge representation	K2	2	Lecture with Discussion	BB/ICT
3		Describe Simple relational knowledge	K2	2	Lecture with Discussion	BB/ICT
4		Describe Inheritable knowledge	K2	2	Lecture with Discussion	BB/ICT

5		Describe Inferential knowledge	K2	2	Lecture with Discussion	BB/ICT
6		Describe Procedural knowledge	K2	1	Lecture with Discussion	BB/ICT
7		Discuss the Issues in knowledge representation	K2	1	Lecture with Discussion	BB/ICT
8		Explain the frame problem	K2	1	Lecture with Discussion	BB/ICT
Total				13		

Total classes:60

Data Mining Lab

Academic Year: 2022-23

Year/ Semester: V

Name of the Course: Data Mining Lab

Programme: B.Tech

Section: A,B,C& D

CourseCode:V20CSL09

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level):

After completion of this course, the students will be able to:

CO1	Demonstrate Data Pre-processing techniques.	[K3]
CO2	Demonstrate Association Rule Mining techniques.	[K3]
CO3	Demonstrate Classification techniques.	[K3]
CO4	Demonstrate the Clustering techniques.	[K3]

Reference Books:

1. Data Mining: Practical Machine Learning Tools and Techniques, Ian H. Witten, Eibe Frank, Mark A. Hall, 3rd Edition, Morgan Kaufmann Publishers.
2. Data Mining Concepts and Techniques, Jiawei Han, Micheline Kamber, Jian Pei, 3rd Edition, Morgan Kaufmann Publishers.
3. Introduction to Data Mining, Pang-Ning Tan, Michael Steinbach, Vipin Kumar, 1st Edition, Pearson Education Inc.

Targeted Proficiency and Attainment Levels (for each course Outcome):

Cos		CO1	CO2	CO3	CO4
Targeted Proficiency Level		75	75	75	75
Targeted level of Attainment	Level 3	70	70	70	70
	Level 2	65	65	65	65
	Level 1	60	60	60	60

Lecture Plan:

S.N O	Course Outcome	Intended Learning Outcomes (ILO)	Knowled ge Level of ILO	No. of Hours required	Pedagogy	Teachin g aids
1	CO 1	Demonstrate Data Preprocessing on predefined Weka dataset labor.arff	K3	3	Lecture With Discussion	ICT
2		Create a student.arff dataset and Demonstrate Data Preprocessing on it	K3	3	Lecture With Discussion	ICT
Total				6		

S.No	Course Outcome	Intended Learning Outcomes(ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Demonstrate Association rule process on predefined Weka dataset contactlenses.arff using apriori algorithm.	K3	3	Lecture With Discussion	ICT
2		Create an employee.arff dataset and demonstrate Association rule process on it using apriori algorithm	K3	3	Lecture With Discussion	ICT
Total				6		

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Demonstrate Classification process on student.arff dataset using j48 algorithm	K3	3	Lecture With Discussion	ICT
2		Demonstrate Classification process on employee.arff dataset using id3 algorithm	K3	3	Lecture With Discussion	ICT
3		Demonstrate Classification process on employee.arff dataset using Naïve Bayes algorithm	K3	3	Lecture With Discussion	ICT
4		Create a customer.arff dataset and demonstrate Classification process on it using j48 algorithm	K3	3	Lecture With Discussion	ICT
Total				12		

S.N o	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Demonstrate Clustering process on predefined Weka dataset iris.arff using simple k-	K3	3	Lecture With Discussion	ICT

		means algorithm				
2		Demonstrate Clustering process on dataset student.arff using simple k- means algorithm	K3	3	Lecture With Discussion	ICT
Total				6		

Total No. of Classes: 30

Web Technologies Lab

Academic Year: 2022-23

Year/ Semester: V

Name of the Course: Web Technologies Lab

Programme: B.Tech

Section: A,B,C& D

CourseCode:V20CSL10

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level):

After completion of this course, the students will be able to:

CO1	Illustrate the basic concepts of HTML and CSS.	K2
CO2	Illustrate Extensible markup language and XML parsers.	K3
CO3	Develop web applications using JDBC.	K3
CO4	Build database driven web applications using JSP.	K3
CO5	Illustrate the basic concepts of Angular and NODE JS.	K2

Reference Books:

1. Web Technologies: HTML, JavaScript, PHP, Java, JSP, XML and AJAX, Black book, Dream Tech.
2. An Introduction to Web Design, Programming, Paul S Wang, Sanda S Katila, Cengage Learning.
3. Web Technologies, 1st Edition 7th impression, Uttam K Roy, Oxford, 2012.

Targeted Proficiency and Attainment Levels (for each course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5
Targeted Proficiency Level		75	75	75	75	75
Targeted level of Attainment	Level 3	70	70	70	70	70
	Level 2	65	65	65	65	65
	Level 1	55	55	55	55	55

Lecture Plan:

S.N O	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours required	Pedagogy	Teaching aids
1	CO 1	Dissemination of Department Vision, Mission, PEOs, POs, PSOs	-	-	-	
		Design HTML fundamental constructs. (i) Headings (ii) Links (iii) Paragraph (iv) Images (v) Tables	K2	3	Lecture With Discussion	ICT
		Design HTML fundamental constructs. (i) Frames (ii) Forms and HTML controls	K2	3	Lecture With Discussion	ICT
3		Design Cascading style sheets (i) Internal (ii) External (iii) Inline	K2	3	Lecture With Discussion	ICT
Total				9		

S.No	Course Outcome	Intended Learning Outcomes(ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Write an XML file which will display the Book information which includes the following: (i) Title of the book (ii) Author Name (iii) ISBN number (iv) Publisher name (v) Edition (vi) Price (a) Write a Document Type Definition (DTD) to validate the above XML file. (b) Write a XML Schema Definition (XSD)	K3	3	Lecture with Discussion	ICT
Total				3		

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Create JSP to insert, delete, and update the details of student into the database	K3	3	Lecture with Discussion	ICT

		using JDBC connectivity.				
Total				3		

S.N o	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids															
1	CO 4	Create a simple JSP to print the current Date and Time	K3	3	Lecture with Discussion	ICT															
2		Develop JSP program calculates factorial values for an integer number, while the input is taken from an HTML form.	K3	3	Lecture with Discussion	ICT															
3		Develop JSP program shows a Sample Order Form A Sample Order Form <table><tr><td>Item</td><td>Price</td><td>Quantity</td><td>Total Price</td></tr><tr><td>DVD</td><td>19.99</td><td>2</td><td>39.98</td></tr><tr><td>CD</td><td>12.99</td><td>9</td><td>116.91</td></tr><tr><td>Diskette</td><td>1.99</td><td>24</td><td>47.76</td></tr></table>	Item	Price	Quantity	Total Price	DVD	19.99	2	39.98	CD	12.99	9	116.91	Diskette	1.99	24	47.76	K3	3	Lecture with Discussion
Item	Price	Quantity	Total Price																		
DVD	19.99	2	39.98																		
CD	12.99	9	116.91																		
Diskette	1.99	24	47.76																		
Total				9																	

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Design a simple Angular JS form	K2	2	Lecture with Discussion	ICT
2		Design a simple Node JS application	K2	1	Lecture with Discussion	ICT
Total				3		

Total No. of Classes: 39

Professional Communication Skills - III

Academic Year: 2022-23

Year/ Semester: V

Name of the Course: Professional Communication Skills - III

Programme: B.Tech

Section: A,B,C& D

CourseCode:V20ENT04

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level):

After completion of this course, the students will be able to:

CO1	Distinguish the subtle meanings of various words in different contexts, recognize similar words as well as words with contrast meanings and use them appropriately. Express writer's tone and relevant ideas using different types of writing skills and prepare a resume to showcase skills and accomplishments. Organize thoughts in the discussions and express views without reticence. Develop the ability to write different types of essays in a structured way, maintaining cohesion and logic.	[K4]
CO2	Identify the central theme and arrange the scrambled sentences into a meaningful passage. Draft emails with appropriate subject-lines and relevant content. Compare different pairs of words, recognize the relationship between the head words and the options to siphon correct analogy Choose an appropriate word to make a sentence meaningful. Infer the meaning of the picture by thinking out of the box and speak without inhibitions and face interviews with aplomb.	[K2]
CO3	Analyze appropriate methods of logical thinking on Ratio and Proportion, Partnership, LCM and HCF, Number System, Areas & Volumes.	[K4]
CO4	Demonstrate problem solving skills through the concepts of Percentages, Profit and loss, Simple Interest & Compound Interest and Allegation.	[K3]
CO5	Calculate the end results of Cubes, Dice and Data Analysis, Time & Work, Time & Distance, Race & Games.	[K4]

Text Books:

- 1: PIC-VOC
- 2: VERBAL ABILITY-2
- 3: WORK BOOK ON APTITUDE

Reference Books:

1. Dr.Sujani Tata et al., Pic Voc (2015) – Published by Sri Vasavi Engineering College
2. Lewis Norman, Word Power Made Easy (2008). Goyal Publishers & Distributors Pvt. Ltd.
3. Dr.Shalini Verma, Reetesh Anand, Word Power Made Handy(2017). S Chand Publications.

4. *R S Aggarwal, Objective General English (2017). S Chand Publications.*
5. Sunita Mishra & C.Muralikrishna, Communication Skills for Engineers (2006). Dorling Kindersley (India) Pvt. Ltd., licensees of Pearson Education in South Asia.
6. Charles W Hanson. Resume: Writing 2020 The Ultimate Guide to Writing a Resume that Lands YOU the Job! (2019).
7. Raymond Murphy. Essential Grammar in Use (1985).Cambridge University Press
8. Seely John. The Oxford Guide to Writing & Speaking (2004). Oxford University Press.
9. Jain,T.S. & Gupta. , 2010, Interviews and Group Discussions, Upkar's Publications.
10. Training & Placement cell, 2020, Workbook -1 on Aptitude, Sri Vasavi Engineering College.
11. M Tyra, 2013, Magical Book on Quicker maths, BSC Publications.
12. K Kundan & M Tyra, 2009, Practice Book on Quicker Maths, BSC Publications.
13. Dr. RS. Agarwal , 2017, Quantitative Aptitude, Sultan Chand Publications
14. Dr. RS. Agarwal, 2017, A modern approach to verbal & on verbal reasoning, Sultan Chand Publications.

Hyperlinks

1. <https://www.indiabix.com/>
2. <https://www.campusgate.co.in/>
3. <https://www.questionpaper.org/>

Targeted Proficiency and Attainment Levels (for each course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
1	50	60
2	50	60
3	50	60
4	50	60
5	50	60

Lecture Plan:

S.N O	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours required	Pedagogy	Teaching aids
1	CO 1	Infer the contextual meaning of words and its contextual usage.	K2	4	Lecture	PPT/A.V
2		Predict the Synonyms-Antonyms of words.	K2	4	Discussion	A.V
3		Develop their resume as per job description	K3	2	Lecture	PPT/A.V
4		Outline paragraphs and essays	K4	2	Lecture	A.V
5		Arrange their ideas logically and present effectively in GDs	K4	2	Lecture	PPT/A.V
6		Connect their views logically and coherently to participate in JAM, and Presentations	K4	3	Lecture	PPT/A.V
7		Relate their creative pursuit to design Advertisements	K4	1	Lecture	PPT/A.V
Total				18		

S.No	Course Outcome	Intended Learning Outcomes(ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Recognize and master “Tenses’ and ‘Voice’ to speak and write effectively.	K1	2	Lecture	PPT/A.V
2		Present their ideas concisely to draft Emails	K1	2	Lecture	PPT/A.V
3		Explain an unfamiliar concept or idea using an analogy.	K2	2	Lecture	PPT/A.V
4		Express their achievements and SWOT confidently to face different types of Interviews	K2	2	Lecture	PPT/A.V
5		Interpret the given picture and write a creative paragraph using writing strategies	K2	2	Lecture	PPT/A.V
Total				10		

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Identify the Next letter or Number in a correct Relation.	K4	2	Lecture	PPT/A.V
2		Justify the relation between words and Numbers.	K4	2	Lecture	PPT/A.V

3		Identify different one from group of terms.	K4	2	Lecture	PPT/A.V
4		Describe their Rank in a class or in a Computation.	K4	2	Lecture	PPT/A.V
Total				8		

S.N o	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Classify the ages in family members/ Explain the relation between numbers.	K3	2	Lecture	PPT/A.V
2		Calculate the Actual time in Mirror and Water/ Classify the Images.	K3	2	Lecture	PPT/A.V
3		Differentiate the logic behind the conclusions.	K3	2	Lecture	PPT/A.V
4		Explain the logic for a given problem.	K3	2	Lecture	PPT/A.V
Total				8		

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Choose the correct relation between the persons.	K4	2	Lecture	PPT/A.V
2		Show the correct direction.	K4	2	Lecture	PPT/A.V
3		Calculate the Average of data.	K4	2	Lecture	PPT/A.V
4		Relate the correct day for a given date and angle between two hands of a clock.	K4	2	Lecture	PPT/A.V
5		Intercept data.	K4	2	Lecture	PPT/A.V
6		Report the Real Time Scenarios possibility..	K4	2	Lecture	PPT/A.V
Total				12		

Total No. of Classes: 56(28+28)